

ZIG Gravity Sandbox — Master Implementation Roadmap

v5.0 (Final Unified)

Lineage: v1/v2 (primary research line, workspace-verified) → v3.0 (first counter-analysis, merged into v4) → v4.0 → **two independent external audits of v4** (Auditor A: 3 findings; Auditor B: 18 findings), every one of which was re-derived by computation before acceptance. This document supersedes all prior versions and is the single architectural authority. Gemini implements against this document and makes no architectural decisions.

Implementing agent: Gemini, Google AI Studio Build mode → Android (Kotlin + Jetpack Compose, single-activity, single-module, no NDK). **Typography policy (owner directive):** the app already has its font. **Gemini uses the app's existing default font everywhere and adds no font dependencies or typography changes.**

PART I — SECOND-ROUND ACCURACY AUDIT (verdicts computed, not judged)

I.A — Auditor A (3 findings)

#	Claim	Computed verdict	Disposition
A1	E2 suppression used the wrong exponent. Force suppression is $[r^2/(r^2+\epsilon^2)]^{3/2}$, not $r^2/(r^2+\epsilon^2)$ — v4's "6.34%" should be ~9.3%	CORRECT — accepted. Softened $a = GM/r/(r^2+\epsilon^2)^{3/2}$ vs true $GM/r^2 \rightarrow$ ratio = $(r^2/(r^2+\epsilon^2))^{3/2}$. At $\epsilon=10^8$, $r=3.844\times 10^8$: $(0.9366)^{1.5} = 0.9064 \rightarrow 9.36\%$ weakening (worse than everyone — v3.0 said 6.3%, v4 endorsed it). Also cascades: at 10^9 m the $\epsilon=10^8$ force error is 1.48% (v4 said 0.99%); with $\epsilon=10^6$ the force distortion at the real Moon distance is 1.0×10^{-5} ($\approx (3/2)(\epsilon/r)^2$, exact 1.015×10^{-5}). One pedantic note: $r^2/(r^2+\epsilon^2)$ is not itself the potential ratio either — the Plummer potential ratio carries exponent $1/2$; the substance of the finding stands	All E2 numbers corrected throughout (§3.2, §3.6a, Part I record). The conclusion ($\epsilon_{\text{soft}} = 10^6$ m) is strengthened, not changed
A2	E4 worst case is T $\approx 64,810$ s (18 steps/orbit), not 303,660 s (84) — computed with $r = 1.122\times 10^{10}$ m	Arithmetic correct; radius assumption not legal — v4's real fault was omitting the size table. Their $r = 1.122\times 10^{10}$ m is the marble–marble contact distance (10 dp). Two 50 M_SUN black holes are BLACK_HOLE bodies whose rings are 14 dp (default) $\rightarrow$ contact 28 dp = 3.142×10^{10} m $\rightarrow T = 303,659$ s = 84.3 steps/orbit (v4's figure). Even at the minimum legal BH ring (10 dp, contact 20 dp) $T = 183,313$ s = 50.9 steps/orbit . Their 18-step case requires 100 M_SUN at a 10-dp contact — impossible under type/mass rules. However: v4 never stated the dp table, so the confusion was v4's fault	Size table reinstated and made normative (§3.6a); worst case documented as a three-case set: default 84.3 $\rightarrow$ 675, legal-min 50.9 $\rightarrow$ 407, unreachable floor 18.0 $\rightarrow$ 144 (kept for honesty, labeled illegal)
A3	BH + BH collision is a logical paradox — "hole never the absorbed side" cannot resolve when both bodies are holes	CORRECT — genuine spec gap (v4 dropped v2's survivor-inheritance sentence)	Deterministic survivor rule added (§3.7): in any merge, the larger mass survives; equal masses $\rightarrow$ lower array slot survives; a BLACK_HOLE always survives over a non-hole regardless of mass ordering... exception: if both are holes, larger-mass rule applies first. Ring radius grows by volume conservation; r_s recomputes; BodyMerged emitted (BH_BH subtype for teaching). New test 40 bhBhMergeDeterministicSurvivor

I.B — Auditor B (18 findings)

#	Claim	Verdict	Disposition
B1	Earth-year 31.6 s figure	No error (auditor self-confirmed)	—
B2	30 fps 16 $\times$ substeps "should be 74.07, not 148"	REJECTED — auditor's own arithmetic error. They divided the 60-fps sim-seconds (2.667×10^5) by 3600 and labeled it 30 fps. At 30 fps the frame is twice as long: $(1/30)\cdot 10^6\cdot 16 = 5.333\times 10^5$ sim-s $\rightarrow$ 148.1 substeps $\rightarrow$ capped at 96 $\rightarrow$ effective $16\cdot(96\cdot 3600\cdot 30/10^6)/16 = 10.37\times$. v4 was correct	No change; the correct derivation is now shown inline in §3.6b so this never resurfaces
B3	Moon-distance distortion " 6.8×10^{-6} , not $\sim 3\times 10^{-6}$ "	Half-right, superseded by A1. Their 6.8×10^{-6} is the potential-form error (v4's $\sim 3\times 10^{-6}$ was simply wrong); the physically relevant force error is 1.0×10^{-5}	Corrected via A1

#	Claim	Verdict	Disposition
B4	Refinement slowdown at 16× not quantified ("~10.4×")	Point adopted; their number wrong. With all steps refined at depth 3: needed inner steps at 16×/60 fps = $74.1 \times 8 \approx 593 \gg 96$. The cap then delivers $96 \cdot (3600/8) \cdot \text{fps sim-s/s} = \mathbf{2.59 \times 10^6 \text{ at 60 fps} \rightarrow \text{effective floor} \approx 2.6 \times \text{ at any speed setting}$ (1.30×10^6 at 30 fps $\rightarrow \approx 1.3 \times$). At 1× only $4.63 \times 8 = 37.0$ inner steps are needed $\leq 96 \rightarrow \text{full 1× rate preserved}$. Their 10.4× figure was the no-refinement 30 fps cap case	Slowdown floor table added (§3.6c)
B5	Threshold 0.2 unjustified; recursion re-check unspecified; depth-3 sufficiency unproven	Adopted as spec clarification. Semantics now explicit: evaluate trigger $\rightarrow$ while (violation $\wedge$ depth<3) halve and re-evaluate $\rightarrow$ at depth 3 execute regardless (bounded best-effort; integrity layers protect; adequacy enforced empirically by tests 36–37 on the crafted worst case, not by proof). 0.2 rationale: a step consuming $\leq 20\%$ of pair separation bounds close-encounter local error; it is a tunable numerical parameter like ϵ/DT , gated by tests	§3.6c rewritten
B6	Refinement slows the whole sim, not just the pair	Same as B4 — adopted, documented (global-step refinement is intentional: deterministic, simple, and momentary in practice)	§3.6c
B7	Wormhole cooldown spatial-only is insufficient	CORRECT — adopted. A body exiting at 1000 km/s crosses the 1.5R spatial gate in $\sim 4,488 \text{ sim-s} \approx \mathbf{4.5 \text{ ms wall at 1×}} \rightarrow$ rapid legal shuttling. Fix: dual cooldown — spatial (fully exit 1.5× partner ring) AND temporal ($5 \times 10^5 \text{ sim-s}$ per body $\rightarrow \geq 0.5 \text{ s wall at 1×}$, 31 ms at 16×)	§3.13; test 25 extended to assert the inter-traversal gap
B8	"Separations below $\sim 10^{10} \text{ m}$ don't exist pre-contact" misleading for mixed sizes	Adopted. Tight statement: the minimum legal contact is the two smallest radii (3 dp asteroid + 4 dp marble) = 7 dp = $7.854 \times 10^9 \text{ m}$. All margins restated against this	§3.6a corrected (margins: 100 km/s $\rightarrow 21.8 \times$; BH-graze $\rightarrow 8.7 \times$; 1000 km/s cap $\rightarrow 2.2 \times$)
B9	"min($2 \times \text{escape}$, 1000 km/s)" conflates soft UI guidance and hard engine clamp	Adopted. Their check verified: $2 \times v_{\text{esc}}$ at 10^{10} m from 50 M_SUN = 1152 km/s > cap. Clarified: engine hard clamp = 1000 km/s absolute (clampVelocity) ; UI guidance = slingshot scale caps at $\min(2 \times v_{\text{esc_local}}, 1000 \text{ km/s})$	§3.7, test 38 reworded
B10	8,766 steps/orbit	No error (sidereal year)	—
B11	"Plummer-softened" terminology	No error ($a = Gm \cdot r / (r^2 + \epsilon^2)^{3/2}$ is precisely Plummer)	—
B12	Kepler comparisons are approximate in the softened regime	Adopted as footnote. At test 6's periapsis (0.5 AU): distortion = $(3/2)(10^6/7.48 \times 10^{10})^2 \approx 2.7 \times 10^{-10}$ — negligible by ~ 7 orders vs the 1% tolerance	Footnote in §3.16
B13	Kinematic release resumes with stale acceleration	CORRECT — real Verlet subtlety. Fix: any mutating intent (drag move/release, mass/radius/velocity change, add/remove) calls <code>computeAccelerations()</code> before the next substep, keeping the Verlet <code>a</code> synchronized	§3.11; new test 39 <code>kinematicReleaseUsesFreshForces</code>
B14	Rollback state underspecified	Adopted. Rollback restores: all SoA arrays + simTime + events appended during the failed substep are discarded. The accumulator is not rolled back (it lives in the ViewModel; the debt-discard rule governs it). Consecutive-failure counter $\rightarrow 3 \rightarrow$ auto-pause	§3.8; test 27 extended to assert simTime rollback
B15	Equal-density assumption stated only in passing	Adopted. Consolidated modeling-assumptions box (§3.4)	§3.4

#	Claim	Verdict	Disposition
B16	Preview trajectory may not match refined engine behavior	Adopted and improved — this exposed a real v4 error both auditors missed. Full mutual preview (600 steps × 190 pairs) costs ≈5.7 MFLOP/frame ≈ 3–11 ms on a mid-range phone — v4's "<0.5 ms" claim was wrong. Fix: the preview integrates the candidate body as a test particle in the frozen field of the other 19 bodies (600 steps × 20 sources ≈ 0.5 MFLOP ≈ 0.2–1 ms). Labeled approximate («پیش‌نمایش تقریبی») when the launched body is massive	§3.9 rewritten
B17	Trail length semantics vs speed	Clarified (design confirmed). Per-frame recording yields a constant ~3–4 s observed-motion window at 60 fps — trail length scales with what the user actually sees motion-wise, which is the perceptually correct behavior. Buffer 240 points; rationale documented	§3.9
B18	"distortion < 10 ⁻⁵ for all r ≥ 10 ⁸ m" is false at the boundary	CORRECT — adopted. With the force factor, distortion at 10 ⁸ m = 1.5×10⁻⁴ (their potential-form 10 ⁻⁴ was also low). Replaced by the formula distortion ≈ (3/2)(ε _{soft} /r) ² plus anchor values: 1.5×10 ⁻⁴ @ 10 ⁸ m · 1.0×10 ⁻⁵ @ 3.844×10 ⁸ m · 6.7×10 ⁻¹¹ @ 1 AU	§3.2 corrected

I.C — Errors found in this pass that BOTH auditors missed

1. **Preview cost** (B16 above): v4's "<0.5 ms" was computed for the wrong workload; mutual integration is 3–11 ms/frame. Fixed by the test-particle preview.
2. **BH-graze speed used the wrong contact distance:** v4's "344 km/s, 9× margin" evaluated escape speed at the 10-dp marble contact instead of the actual 19-dp marble+ring pair distance → correct values **250 km/s**, margin 8.7× against the absolute minimum contact (23.7× against the pair's own contact). All no-tunneling conclusions survive with explicit margins.
3. **Sustained-refinement floor:** neither auditor computed it correctly (A said nothing; B said 10.4×): the true floor is **≈2.6× at 60 fps / 1.3× at 30 fps**, and 1× runs at full rate (37.0 ≤ 96 inner steps).

Net changes to v4: E2 numbers globally corrected (exponent ^{3/2}); normative dp size table added; worst case re-expressed as a three-case set; BH+BH survivor rule added; dual wormhole cooldown; rollback/kinematic/trigger semantics made explicit; preview switched to test-particle preview model with honest cost; clamp split into hard engine cap vs soft UI guidance; **tests 39 and 40 added (total 40)**; one auditor claim rejected with arithmetic (B2).

PART II — LOCKED DECISIONS (final, non-negotiable)

#	Decision	Final value
1	Architecture	Single-module Compose app; pure-Kotlin physics core (zero <code>android.*</code> imports, JVM-testable) → ViewModel frame loop → Compose UI; education is a decoupled observer
2	State layout	Structure-of-arrays: fixed-capacity(20) <code>DoubleArray</code> <code>x/y/vx/vy/ax/ay/mass/radius</code> , <code>LongArray</code> <code>ids</code> , <code>ByteArray</code> <code>types</code> , <code>BooleanArray</code> <code>flags</code>
3	Units	Real SI + <code>Double</code> throughout the engine; <code>Float</code> only at the draw boundary; pacing via compression constant, never unit scaling
4	Physics	Exact pairwise Newtonian N-body, $O(n^2)$, $n \leq 20$ (190 pairs), Plummer-softened $\epsilon_{\text{soft}} = 1 \times 10^6 \text{ m}$ (force distortion $\approx (3/2)(\epsilon/r)^2$), all bodies attract all bodies
5	Integrator	Velocity Verlet (KDK) — 2nd-order, symplectic, 1 force-eval/step, synchronized (x,v); <code>computeAccelerations()</code> re-run after every mutating intent
6	Time architecture	Fixed DT = 3600 s + accumulator; BASE = 10^6 sim-s/real-s ; speeds 0.1/0.25/1/4/16× multiply BASE, never DT; MAX_SUBSTEPS = 96 (counts refined inner steps)
7	Collisions	Circle–circle $O(n^2)$; merge by default (mass/momentum/volume-conserving; survivor = larger mass, tie → lower slot; hole beats non-hole); opt-in marble bounce ($e=0.4$); BH+BH resolved deterministically
8	Rendering	Jetpack Compose Canvas only; single composable; imperative draw; draw-phase-only state reads; zero draw-lambda allocation; canvas pinned LTR
9	Threading	Single-threaded inside <code>withFrameNanos</code> on the main dispatcher
10	State bridge	<code>StateFlow<SimSnapshot></code> default; frame-tick + raw arrays permitted only if profiling demands
11	Typography	The app's existing default font. No font dependencies. No typography changes.
12	Black hole	Newtonian point mass (default 5 <code>M_SUN</code> , cap 50 <code>M_SUN</code> , ring 10–20 dp default 14); capture radius = displayed ring radius (one shared constant); self-consistent <code>r_s</code> display
13	Wormhole	Paired massless mouths; velocity-preserving teleport; dual cooldown (spatial $1.5 \times \text{ring exit}$ AND $5 \times 10^5 \text{ sim-s}$) ; always labeled theoretical
14	Education	Persian-first 3-tier cards + 8 POE challenges + prediction micro-prompts; decoupled detectors
15	Dependencies	Zero physics/rendering/font engines: AndroidX + Kotlin stdlib + <code>kotlinx.serialization</code> + <code>DataStore</code>
16	Integrity	3 layers: softening prevents → snapshot rollback (arrays + <code>simTime</code> + failed-step events) repairs → body quarantine (last resort)
17	Edge-case safety	Engine hard velocity cap 1000 km/s (primary) + UI guidance $\min(2 \times v_{\text{esc}}, 1000 \text{ km/s})$ + Adaptive Safety Refinement depth ≤ 3 (secondary, dormant, deterministic; sustained-use floor $\approx 2.6 \times @60 \text{ fps}$)
18	Testing	40 automated JVM tests + 16-item manual UI checklist
19	Delivery	11 phases (0–10); each phase's Gemini prompt generated only after owner approval of the previous

PART III — IMPLEMENTATION SPECIFICATION

3.1 Product definition (locked)

A **2D tabletop gravity experiment**: a premium neutral material surface (dark or light), viewed from above, ≤ 20 marble-like bodies, exact 2D Newtonian N-body mechanics. Depth cues are rendering-only. Not a space simulator, videogame, or scientific workstation.

Priority order: **1) Scientific correctness** **2) Educational value** **3) Interaction quality** **4) Visual clarity** **5) Performance** **6) Aesthetic polish**.

Honesty requirement: preset distances/radii are not to scale (stated once, clearly). Masses, G, and velocities are real, so emergent behavior — periods, escape speeds, energy exchange — is genuinely correct.

3.2 Constants charter (locked in Phase 1; `constantsAreExactSI` test enforces)

Constant	Value	Notes
G	$6.67430 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$	CODATA 2018; never tunable
c	$2.99792458 \times 10^8 \text{ m/s}$	exact; never tunable
M_SUN / R_SUN	$1.989 \times 10^{30} \text{ kg}$ / $6.957 \times 10^8 \text{ m}$	educational rounding
M_EARTH / R_EARTH	$5.972 \times 10^{24} \text{ kg}$ / $6.371 \times 10^6 \text{ m}$	
M_MOON / R_MOON	$7.348 \times 10^{22} \text{ kg}$ / $1.737 \times 10^6 \text{ m}$	
AU	$1.496 \times 10^{11} \text{ m}$	
Earth orbital speed	29.78 km/s	$= \sqrt{GM_{\text{SUN}}/AU}$ ✓
Moon	$3.844 \times 10^8 \text{ m}$ · 27.32 d · 1.022 km/s	reference values
r_s(1 M_SUN)	2954.1 m self-consistent (2953.25 m with true mass)	display "≈ 2.95 km"; self-consistency test only
r_s(1 M_EARTH)	8.87 mm	
ϵ_{soft}	$1 \times 10^6 \text{ m}$	numerical-stability parameter; force distortion $\approx (3/2)(\epsilon_{\text{soft}}/r)^2$: 1.5×10^{-4} @ 10^8 m · 1.0×10^{-5} @ $3.844 \times 10^8 \text{ m}$ · 6.7×10^{-11} @ 1 AU; not UI-exposed
DT	3600 s	numerical-stability parameter; not UI-exposed
BASE	$10^6 \text{ sim-s/real-s}$	Earth year $\approx 31.6 \text{ s}$, Moon month $\approx 2.4 \text{ s}$ at $1 \times$
MAX_BODIES / MAX_SUBSTEPS	20 / 96	96 counts refined inner steps
Scene scale	viewport width = 3 AU	metersPerDp $\approx 1.122 \times 10^9$ (400 dp); derived, not stored
Velocity hard cap	1000 km/s	engine-level <code>clampVelocity</code> ; UI guidance separately at $\min(2 \times v_{\text{esc_local}}, 1000 \text{ km/s})$

3.3 Package structure (single `:app` module)

```
com.zig.gravity/
├─ physics/                ← PURE Kotlin. ZERO android.* imports.
│  ├─ SimArrays.kt         (SoA state, capacity 20)
│  ├─ BodyType.kt          (SUN, PLANET, MOON, ASTEROID, TEST_MARBLE, BLACK_HOLE, WORMHOLE_MOUTH)
│  ├─ NBodyEngine.kt        (force pass, Verlet, refinement trigger, collisions, capture, teleport)
│  ├─ Collision.kt          (merge + impulse-bounce pure functions, survivor rule)
│  ├─ SimEvent.kt           (BodyMerged[+BH_BH subtype], BodyEscaped, OrbitStabilized, OrbitDecayed,
│  │                         TwoBodyDance, BlackHoleCapture, WormholeTraversal, NumericalFailure,
│  │                         BodyAdded, BodyRemoved)
│  └─ EngineConstants.kt    (§3.2 – the charter)
├─ edu/
│  ├─ detectors/            (pure functions over snapshots)
│  ├─ TeachingCatalog.kt    (fa/en card content)
│  └─ Challenges.kt         (8 POE experiments + micro-prompt content)
├─ sim/
│  ├─ SimulationViewModel.kt (frame loop, accumulator, intents, snapshot publish)
│  ├─ SimSnapshot.kt · Presets.kt · SaveState.kt (serialization + DataStore)
└─ ui/
   ├─ theme/                (ZigTheme dark/light – app's existing typography, untouched)
   ├─ TabletopCanvas.kt      (THE single Canvas, pinned LTR)
   ├─ HudBar.kt · InspectorSheet.kt · AddBodySheet.kt · TeachingCard.kt
   └─ MainActivity.kt
```

Binding rule: `physics/` and `edu/detectors/` compile on a bare JVM. This enables all 40 tests without an emulator.

3.4 Physics model + modeling-assumptions box

$a_i = \sum_{j \neq i} G \cdot m_j \cdot (r_j - r_i) / (|r_j - r_i|^2 + \epsilon_{\text{soft}}^2)^{3/2}$ — the acceleration form of $F = G \cdot m_1 \cdot m_2 / r^2$, all 190 pairs per force pass. Earth pulls the Sun; the Moon pulls both (`earthPerturbsSun`, `moonAffectsBoth` prove it forever). Massless bodies (test marbles, wormhole mouths) exert no gravity — falls out of $m_j = 0$, no special case — while fully feeling it.

Modeling assumptions (stated once, referenced everywhere): 1. Gravity is Newtonian only (no relativity is computed anywhere, including black holes). 2. Preset distances and radii are not to scale; masses, G, velocities are real. 3. Mergers assume **equal density** — $r' = (r_1^3 + r_2^3)^{1/3}$ conserves volume and, under equal density, conserves density exactly; real bodies differ (a real merger of a gas giant and a rocky planet would not). 4. Wormhole teleport preserves velocity — a modeling choice for undefined physics. 5. The slingshot preview is a test-particle approximation (see §3.9).

Derived physics for features: circular speed $\sqrt{GM/r}$; escape speed $\sqrt{2GM/r}$; specific orbital energy $\epsilon_{\text{orb}} = v^2/2 - GM/r$ (bound $\Leftrightarrow \epsilon_{\text{orb}} < 0$); eccentricity vector from state vectors. **Collision radius = visual radius in scene meters.** No zoom in v1.

3.5 Integrator — Velocity Verlet (KDK)

```
fun step(dt: Double) {
    for (i in 0 until n) { vx[i] += ax[i]*dt/2; vy[i] += ay[i]*dt/2 } // KICK ½
    for (i in 0 until n) { x[i] += vx[i]*dt; y[i] += vy[i]*dt }      // DRIFT
    computeAccelerations()                                           // softened O(n²)
    for (i in 0 until n) { vx[i] += ax[i]*dt/2; vy[i] += ay[i]*dt/2 } // KICK ½
    simTime += dt
    resolveCollisions(); validateState()
}
```

Selected over Euler (unbounded energy growth), symplectic Euler (bounded but sloppier at identical cost), and RK4 (non-symplectic secular drift, 4× cost). Deterministic, time-reversible, ~15 lines. Earth orbit = 8,766 steps/orbit at DT=3600.

Synchronization rule: every mutating intent (drag move/release, mass/radius/velocity change, add/remove) runs `computeAccelerations()` before the next substep so the first kick never uses a stale acceleration (Auditor B13; test 39).

3.6 Time architecture + scene-scale facts

3.6a — Normative visual-size table and its consequences. Type dp ranges (inspector sliders bounded by these; dp→meters at metersPerDp $\approx 1.122 \times 10^9$):

Type	dp range	Default
ASTEROID	3–6	4
TEST_MARBLE	4–7	5
MOON	5–8	6
PLANET	8–16	10
SUN	20–32	26
BLACK_HOLE ring	10–20	14

Consequences: minimum legal contact distance = 3 dp + 4 dp = 7 dp = **7.854×10^9 m**. Per-substep travel vs that minimum: 100 km/s $\rightarrow 3.6 \times 10^8$ m = **21.8× margin**; the fastest legal near-contact speed (grazing a default 5 M_SUN ring, marble 5 dp + ring 14 dp = 19 dp: $v_{\text{esc}} = 250$ km/s $\rightarrow 9.0 \times 10^8$ m) = **8.7× margin**; the 1000 km/s hard cap $\rightarrow 3.6 \times 10^9$ m = **2.2× margin. No tunneling is possible at any legal speed.** Worst-case orbital resolution (two 50 M_SUN holes, Kepler $T = 2\pi\sqrt{(r^3/G \cdot 100 \text{ M_SUN})}$): default rings (28 dp) $T = 303,659$ s = **84.3 steps/orbit**; minimum rings (20 dp) = 183,313 s = **50.9 steps/orbit**; the oft-quoted 64,811 s (18.0 steps/orbit) requires 100 M_SUN at a 10-dp contact — **unreachable under the type/mass/size rules** (recorded as the illegal floor for honesty).

3.6b — Accumulator (per frame, inside `withFrameNanos`):

```
accumulator += min(frameDeltaSeconds, 0.1) × BASE × speed
while (accumulator >= DT && steps < MAX_SUBSTEPS) { engine.step(DT); accumulator -= DT; steps++ }
if (steps == MAX_SUBSTEPS) accumulator = 0.0 // discard debt, never spiral
publishSnapshot()
```

Speed	Sim-s/frame @60 fps	Substeps @60 fps	Substeps @30 fps
0.1×	1.667×10^3	0.46	0.23
0.25×	4.167×10^3	1.16	0.58
1×	1.667×10^4	4.63	2.31
4×	6.667×10^4	18.5	9.26
16×	2.667×10^5	74.1 ✓ fits under 96	148.1 $\rightarrow$ capped 96 $\rightarrow$ effective 10.37× (30-fps frames are twice as long: $(1/30) \cdot 10^6 \cdot 16/3600 = 148.1$ — not 74.1)

Pause = stop accruing; resume continues; reset = reload preset deep copy. Speed labeled honestly as «سرعت زمان»; G never user-adjustable.

3.6c — Adaptive Safety Refinement (dormant, deterministic, bounded). Before an engine step, compute each pair's predicted closing displacement $s = |v_{\text{rel}}| \cdot dt + \frac{1}{2} \cdot |a_{\text{rel}}| \cdot dt^2$ (free — values exist from the force pass). **Trigger:** any pair with $s > 0.2 \times \text{separation}$. **Semantics:** halve the step, re-evaluate the trigger on the inner step, repeat while (violation $\wedge$ depth < 3); at depth 3 ($dt/8 = 450$ s) execute regardless — bounded best-effort, with the §3.8 integrity layers behind it. Adequacy is enforced empirically by tests 36–37, not by proof; 0.2 joins $\epsilon_{\text{soft}}/DT$ as a test-gated tunable numerical parameter (a step consuming $\leq 20\%$ of separation bounds close-encounter local error). **Charging:** every inner step counts against $\text{MAX_SUBSTEPS}=96$. **Whole-step refinement is intentional** (simple, deterministic; encounters are momentary). **Sustained-refinement floor (all steps refined, worst case):** at 60 fps the cap delivers $96 \cdot 450 \cdot 60 = 2.59 \times 10^6$ sim-s/s $\rightarrow$ **effective $\approx 2.6\times$ regardless of the speed setting** ($1.3 \times 10^6 \rightarrow \approx 1.3\times$ at 30 fps); at 1× only $4.63 \times 8 = 37.0$ inner steps are needed $\leq 96 \rightarrow$ **1× always runs at full rate**. Worst-case resolution with refinement: 84.3→675, 50.9→407 (illegal floor 18.0→144).

3.7 Collision & merger model

Detection per substep: all 190 pairs, $\text{dist} < r_i + r_j$ (scene meters). Response policy:

Pair	Behavior
Any + any (default)	Merge: $m' = m_1 + m_2$ · $v' = (m_1 v_1 + m_2 v_2) / m'$ · $pos' = \text{center of mass}$ · $r' = (r_1^3 + r_2^3)^{1/3}$ (equal-density assumption, §3.4)
TEST_MARBLE + TEST_MARBLE (opt-in, default OFF)	Impulse bounce, $e = 0.4$
Any + BLACK_HOLE	Absorb via the same merge math; a hole always survives over a non-hole
BLACK_HOLE + BLACK_HOLE	Merge with deterministic survivor (Auditor A3): larger mass survives; equal masses → lower array slot survives. Ring grows by volume conservation; r_s recomputes; <code>BodyMerged (BH_BH)</code> emitted
Any + WORMHOLE_MOUTH	No collision; teleport trigger
MOUTH + MOUTH	Never interact

The survivor rule (larger mass, tie → lower slot) applies to **all** merges, eliminating every ambiguous case. Extreme mass ratios need no special case — momentum conservation alone produces "pebble vanishes into star, star barely moves." Bounce: normal impulse $j = -(1+e) \cdot v_n / (1/m_1 + 1/m_2) + 0.8 \times$ -penetration positional split.

Velocity bounds (primary edge-case mitigation): the **engine hard clamp is 1000 km/s absolute** (`clampVelocity()`, pure, tested). Separately, the **UI guidance** scale for slingshot and inspector caps at `min(2 × v_esc_local, 1000 km/s)` (soft guidance; the engine clamp catches anything anyway). Type-aware inspector ranges (§3.6a table, §3.11 mass ranges) bound pathological configurations.

Rejected: Barnes-Hut (20 bodies); continuous collision detection (margins in §3.6a make it pointless); debris spawning (violates the 20-body contract and the calm).

3.8 Numerical integrity — 3 layers

1. **Prevent:** $\epsilon_{\text{soft}} = 10^6$ m keeps forces finite through near-contact.
2. **Repair:** per-substep NaN/Inf scan → **rollback restores all SoA arrays AND simTime; events appended during the failed substep are discarded** (the accumulator is not rolled back — it lives in the ViewModel and is governed by the debt-discard rule) → emit `NumericalFailure`; 3 consecutive rollbacks → auto-pause with a calm notice.
3. **Last resort:** quarantine a repeat-offender body with an event rather than let it poison the pairwise sums.

3.9 Rendering — Compose Canvas (font-agnostic)

One `TabletopCanvas` owns the scene; chrome is ordinary Compose above. State read **only inside the draw lambda** (zero recomposition per frame). **Zero allocation in the draw lambda** — all Brush/Path/PathEffect/TextMeasurer preallocated; static background layers via `drawWithCache`. Shadows: soft radial-gradient ellipses (never `setShadowLayer`). Bodies are marbles: shadow → radial-gradient base → optional rim → restrained specular → selection ring → cached label. No rotation, surfaces, atmospheres, terminators. `Double→Float` exactly once, at the draw boundary. **The canvas is pinned LTR** (`CompositionLocalProvider(LocalLayoutDirection provides Ltr)`); **only chrome mirrors under RTL**. Typography: the app's existing default font — no font resources, no font dependencies, no typography edits.

Trails: preallocated ring buffers, **240 points, one point recorded per rendered frame** → a constant ~4 s observed-motion window at 60 fps; trail extent scales with visible motion (perceptually correct at every speed). Drawn as low-alpha stroked paths with 2–3 stepped-alpha segments.

Slingshot preview (corrected, Auditor B16): while dragging, integrate the candidate body as a **test particle through the frozen field of the other bodies** (fixed DT, up to 600 steps or until off-screen/collision) ≈ 0.5 MFLOP ≈ 0.2 – 1 ms per frame. Full mutual integration (≈ 5.7 MFLOP ≈ 3 – 11 ms) is **forbidden** — it would blow the frame budget. The preview is labeled approximate («پیش‌نمایش تقریبی») whenever the launched body's mass exceeds 1% of system mass; for test marbles it is essentially exact.

Visual direction ("precision instrument on a desk," never space):

	Dark	Light
Tabletop	Warm charcoal <code>#16181D</code> → <code>#1C1F26</code> , top-light vignette	Warm paper <code>#F4F1EA</code> → <code>#E9E4D9</code> , faint grain vignette
Bodies	Desaturated premium tones (Sun ivory-amber, Earth slate-blue, Moon bone, asteroid warm grey, marble porcelain)	Same, slightly deepened
Accent	Muted brass <code>#D4A853</code>	Deep teal <code>#2F6B63</code>

Dark		Light
Black hole	Near-black #0A0A0C + thin accent ring	Charcoal #26262B + ring
Chrome	Translucent tonal glass, generous spacing	Same, lighter

Forbidden: stars, nebulae, screen shake, particle explosions, accretion renderers, lensing (trails are evidence and never bend), videogame anything.

3.10 GitHub references — verified table (reference only; zero code imported)

Repo	Verified status (fetched directly x3)	Take	Forbid
igndev07/ gravityforge (MIT)	Exists; 1 commit 5034caa, May 28 2026 (3 months past as of 2026-08-31); Kotlin 2.0 + Compose; Barnes-Hut $\theta=0.9$; Velocity Verlet; MVVM + StateFlow	Package layout; Compose-Canvas N-body pattern proof; gesture-table concept	Barnes-Hut; starfield aesthetic; background physics thread; $+\epsilon$ softening (dimensionally wrong); committed build dirs
KPhysics/KPhysics	Real; pure-Kotlin rigid-body engine; license file present but unverified — copy no code	Study impulse/restitution math (clean-room)	As a dependency (dormant, wrong paradigm)
joaomcl/boks2d (MIT)	Real; KMP bindings for native Box2D	Nothing	Everything — NDK impossible in AI Studio; wrong model
ccampo133/ NBodyJS	Real; Kotlin N-body, Verlet + softening documented	Verlet-loop + softened-sum shape; fling-launch concept	kotlin-js plumbing; G=1 toy units
AntonisPsarras/ Aether-Gravity	Real; TypeScript; ships PHYSICS_AUDIT.md	The written-physics-audit culture (this spec's test suite)	Code; product direction
alexvanyo/ composelife (Apache-2.0)	Real; Compose simulation app	Testing culture for sim engines	AGSL/OpenGL renderers; KMP complexity
jBox2D / libGDX / KTX / Filament	Real	Nothing	Wrong paradigm / platform / dimensionality

The architecture is 100% self-sufficient and requires none of these. No unverified repo may justify any deviation from this document.

3.11 Interaction model

Gesture	Action
Tap body / empty	Select (ring + haptic) / deselect
Drag body	Kinematic while held (skips integration); velocity tracked from pointer history; release carries the throw; <code>computeAccelerations()</code> runs on every drag update and on release
Drag outward from body, hold	Slingshot: drag vector = -velocity, live dotted test-particle trajectory preview , release launches (UI scale capped at $\min(2 \times v_{\text{esc}}, 1000 \text{ km/s})$; engine clamps at 1000 km/s)
Long-press body	Inspect · Duplicate · Remove
Long-press empty / HUD "+"	Add catalog (20-cap, friendly notice: «زمین آزمایش پر است — ۲۰ جسم بیشتر جا نمی‌شود»)

Inspector (contextual bottom sheet): mass log-slider (1 = جرم زمین, kg secondary; type ranges: planet 0.01–100 M_EARTH · star 0.1–10 M_SUN · BH 1–50 M_SUN · asteroid/marble $\leq 10^{-3}$ M_EARTH) · visual size slider **bounded by the §3.6a dp table** (real radius info-line) · velocity readout + direction dial + «مدار دایره‌ای کن» ($v(GM/r)$ perpendicular to the dominant attractor) · **prediction micro-prompt strip** («پیش‌بینی‌ات چیست؟»), teaching mode only, once per property per session, never blocks the slider). HUD: play/pause · $\frac{1}{10} \times \frac{1}{4} \times 1 \times 4 \times 16 \times$ · reset · trails · teaching · theme · language · add. Nothing else persistent.

3.12 Black hole

Newtonian point mass (default 5 M_SUN, slider 1–50 M_SUN, ring 10–20 dp default 14) in the exact N-body system — no special force code. **Capture radius = displayed ring radius, one shared constant** (test-enforced). Inspector shows the self-consistent $r_s = 2GM/c^2$ (5 M_SUN → 14.77 km; display «افق رویداد واقعی: $\approx 14/8$ کیلومتر»). Teaching card: gravity here is Newtonian everywhere; the horizon is a GR concept shown as the boundary of no-return; the drawn ring is a display convention. BH+BH mergers emit `BodyMerged(BH_BH)`; the tier-3 card notes that real black-hole mergers radiate gravitational waves — a relativistic effect this Newtonian sandbox does not compute. Visual: matte near-black disk, one thin accent ring, faint static halo, optional slow alpha shimmer. No lensing of any kind.

3.13 Wormhole

Paired massless mouths; no gravity; teleport on center-entry: exit placed just outside the partner along entry direction; **velocity preserved unchanged** (documented modeling choice — the real geometry of a wormhole mouth is undefined physics; orientation transforms degenerate to identity under auto-facing). **Dual cooldown (Auditor B7)**: a body must (a) fully exit 1.5× the partner's ring radius **and** (b) wait 5×10^5 sim-s (≥ 0.5 s wall at 1×) before either mouth can trigger for it again. Labeled «کرم‌چاله (فرضی)» on the object and everywhere. Tier-3 card: Einstein–Rosen bridges are a legitimate mathematical solution; no traversable wormhole has ever been observed; known solutions require exotic matter. Visual: two flat rings, paired warm/cool accents, gentle synchronized pulse.

3.14 Education (Persian-first)

3-tier cards («چه اتفاقی افتاد؟» → «چرا؟» → «بیشتر بدانیم»), opt-in, never blocking. Tone: friendly, intelligent, beautiful — never childish, never stiff. 7 detectors (OrbitStabilized [$\epsilon_{orb} < 0 \wedge \geq 300^\circ$ sweep], BodyEscaped, BodyMerged, BlackHoleCapture, WormholeTraversal, OrbitDecayed, TwoBodyDance), pure Kotlin over snapshots, ≥ 2 s hysteresis. 8 POE challenges: mass-doubling · distance-vs-period · escape velocity · binary barycenter · collision momentum · black-hole capture · wormhole traversal · "why doesn't the Moon fall?" (first-run tutorial). Persian digits ۰–۹ default, Latin fallback setting. Locked glossary: گرانش · مدار · جرم · شعاع · سرعت · سرعت گریز · تکانه/پایستگی تکانه · انرژی جنبشی/پتانسیل · برخورد · ادغام · سیاه‌چاله · افق رویداد · کرم‌چاله · آزمایش · پیش‌بینی · شبیه‌سازی · جسم آزمایشی · سیارک. Content in `TeachingCatalog` (fa/en Kotlin maps); chrome strings in standard res/values + values-en.

3.15 Performance budget

Item	Target
Frame total	≤ 16.6 ms
Physics (≤ 96 substeps × 190 pairs + rare refinement)	< 1.5 ms
Canvas draw (≤ 150 ops) + preview (test-particle)	< 4 ms combined
Recomposition/frame	0
Draw-lambda allocations	0

SoA arrays; drawWithCache statics; cached brushes; trails sampled once/frame; Choreographer-based ms overlay behind a debug flag; 20-body stress preset; optional JankStats in Phase 10.

3.16 Testing — 40 automated JVM tests + manual checklist

Orbital mechanics & integrator: 1 `twoBodyCircularOrbitClosesAfterOnePeriod` (1 AU, 29.78 km/s, $< 1\%$ r) · 2 `totalEnergyDriftBounded` ($< 10^{-3}$ over 10 orbits) · 3 `totalMomentumConservedInFreeFlight` · 4 `earthPerturbsSun` · 5 `moonAffectsBoth` · 6 `eccentricOrbitPeriodMatchesKepler` ($e=0.5$, $a=1$ AU → periapsis 0.5 AU, softening distortion there $\approx 2.7 \times 10^{-10}$ — 7 orders below tolerance) · 7 `hyperbolicFlybyEnergyConserved` **Timestep & determinism:** 8 `substepGroupingInvariant` (bitwise) · 9 `speedEquivalence16xVs1x` (bitwise) · 10 `pauseResumeStateIdentical` · 11 `resetRestoresPresetExactly` · 12 `deterministicReplay` (state hash) · 13 `accumulatorNeverExplodes` **Collisions:** 14 `mergeConservesMomentum` · 15 `mergeConservesMass` · 16 `mergeVelocityFormula` · 17 `mergeRadiusVolumeConserving` · 18 `extremeMassRatioAbsorb` · 19 `bounceImpulseMath1D` · 20 `bounceMomentumConserved2D` · **40** `bhBhMergeDeterministicSurvivor` (equal-mass tie → lower slot; larger mass wins; ring volume-conserving; momentum conserved) **Special bodies & integrity:** 21 `blackHoleCaptureConservesMomentumAndGrowsMass` · 22 `blackHoleRingRadiusIsCaptureRadius` · 23 `schwarzschildRadiusSelfConsistent` ($r_s == 2GM/c^2$ from the same

constants, rel. 10^{-9} ; sanity band 2900–3000 m) · 24 `wormholeTeleportPreservesVelocity` · 25
`wormholeCooldownPreventsOscillation` (asserts inter-traversal gap $\geq 5 \times 10^5$ sim-s per body AND spatial gate) · 26
`bodyCapEnforcedAt20` · 27 `nanGuardRollsBack` (asserts arrays AND simTime restored exactly) · 28
`softeningPreventsSingularity` · 29 `testParticleExertsNoGravity` · 30 `kinematicDragBypassesIntegration` · 39
`kinematicReleaseUsesFreshForces` (first post-release kick uses accelerations computed at the release position) **Foundation:**
31 `constantsAreExactSI` · 32 `saveRestoreRoundTripIdentical` **Detectors:** 33 `orbitDetectorFiresOnClosedSweep` · 34
`escapeDetectorFiresOnUnbound` **Edge-case hardening:** 35 `softeningDoesNotDistortKnownOrbits` (circular speed within
0.1% of unsoftened Kepler at 1 AU and at 3.844×10^8 m; actual distortion $1.0 \times 10^{-5} = 0.001\%$) · 36 `tightOrbitDoesNotDiverge`
(twin 50 M_SUN BHs at contact: energy bounded over 10^4 steps with refinement) · 37 `fastFlybyDoesNotTunnel` (marble at
1000 km/s past a 50 M_SUN BH: merges or deflects, never passes through) · 38 `velocityBoundsClamped` (engine clamp $\leq$
1000 km/s absolute; UI guidance function returns $\min(2 \times v_{\text{esc}}, 1000 \text{ km/s})$)

Manual UI checklist (16 items, Phase 10): 60 FPS feel @20 bodies+trails · gesture feel · slider changes don't recompose canvas
· Persian RTL correct **while canvas stays LTR** · Persian digits · both-theme contrast of every body type · card readability ·
backgrounding pauses · rotation preserves experiment · 20-cap message · reset from any state · calm merge feedback ·
trajectory preview tracks finger · honest speed labels · black hole visible on light theme · velocity caps enforced in slingshot +
inspector.

3.17 Gemini guardrails (verbatim in every phase prompt)

MUST NOT: import any physics/rendering engine · introduce Barnes-Hut or spatial trees · scale `DT` for speed changes · run
physics on a background thread · exceed 20 bodies · make G user-adjustable · expose $\epsilon_{\text{soft}}/\text{DT}/\text{MAX_SUBSTEPS}$ in the UI ·
make the simulation 3D · write planetary shaders/rotation/atmospheres/terminators · add starfields/screen-shake/debris/
particles/lensing · expand scope beyond the tabletop concept · alter §3.2 constants · use `Float` inside the physics engine ·
add, remove, or change any font or typography — use the app's existing default font · mirror the physics canvas under RTL ·
make the slingshot preview a full mutual integration (test-particle only). **MUST:** keep `physics/` and `edu/detectors/` free of
`android.*` imports · keep all 40 tests passing and add each phase's new tests · recompute forces after every mutating intent ·
Persian-first copy with English secondary · honor every rendering/visual restraint above.

PART IV – ROADMAP (11 phases, final)

Workflow: you approve each phase (paste result/screenshots/test output) → I generate that phase's complete, self-contained Gemini prompt (scope, exact spec excerpts, guardrails block, acceptance tests).

Phase	Title	Key builds	New tests
0	Foundation & tabletop shell	Package skeleton; ZigTheme (dark) using the app's existing typography, unchanged ; static Canvas background + vignette; HUD placeholder; EngineConstants with §3.2 values	—
1	Data model, units & state	SoA arrays (capacity 20); BodyType ; add/remove/cap; units charter; code-comments distinguishing physical constants from tunable numerics	31, 26
2	Physics core (headless)	Softened force pass ($\epsilon_{\text{soft}}=10^6$ m); Velocity Verlet; accumulator; clampVelocity (1000 km/s); Adaptive Safety Refinement (depth ≤ 3, test-gated) ; minimal Sun–Earth preset; event queue	1–9, 12, 13, 28, 29, 35–38
3	Rendering & time controls	TabletopCanvas (LTR-pinned); marbles + shadows; trails (240-pt ring buffers); selection ring; frame loop; snapshot flow; HUD play/pause/speed/reset	10
4	Interaction	Tap/drag (kinematic + throw + force recompute)/long-press; inspector (mass/size per §3.6a dp table, velocity, orbit-helper, bounds wiring); add catalog with cap; slingshot + test-particle live preview; prediction micro-prompt strip	30, 39
5	Collisions & mergers	Merge math + events + calm visuals; deterministic survivor rule (incl. BH+BH tie-break) ; opt-in marble bounce	14–20, 40
6	Presets & persistence	6-experiment catalog; autosave/restore; lifecycle hardening; deep-copy reset	11, 32
7	Black hole	Point-mass body; ring=capture shared constant; self-consistent r_s display; honesty card; BH+BH handled per §3.7	21–23
8	Wormhole	Paired mouths; teleport + dual cooldown ; «فرضی» labeling; model-choice card	24–25
9	Educational system	7 detectors; 3-tier cards; 8 POE challenges; micro-prompts; glossary	33–34
10	Light theme, perf & QA	Light tabletop; perf overlay + stress preset; full regression; 16-item manual checklist; final guardrail audit	all 40 green

Sequencing: units/state before physics; physics before pixels; rendering+time together; interaction before collisions; special bodies after core mechanics; education last-but-largest; hardening final. Phases 1–2 deliberately thin — small phases keep an AI coding agent honest.

PART V — VERIFICATION RECORD & REFERENCES

This round (v4 → v5), all computed: force-suppression exponent $3/2$ ($\epsilon=10^8$ @Moon → 9.36% weak; @ 10^9 m → 1.48%; $\epsilon=10^6$ anchors 1.5×10^{-4} @ 10^8 m, 1.02×10^{-5} @ 3.844×10^8 m, 6.7×10^{-11} @1 AU) · twin-50-M_SUN periods 303,659 s (28 dp) / 183,313 s (20 dp) / 64,811 s (illegal 10-dp floor) · 30-fps $16\times = 148.1$ substeps (Auditor B's 74.07 refuted — 60-fps sim-seconds mislabeled) · no-tunnel margins $21.8\times$ / $8.7\times$ / $2.2\times$ vs the 7-dp minimum contact · sustained-refinement floor 2.59×10^6 sim-s/s @60 fps (1.30×10^6 @30 fps); $1\times$ needs 37.0 inner steps ≤ 96 (full rate) · graze speed 250 km/s at the correct 19-dp pair distance (v4's 344 km/s used 10 dp — corrected) · preview costs: test-particle 0.5 MFLOP (~ 0.2 –1 ms) vs mutual 5.7 MFLOP (3–11 ms, forbidden) · spatial-only wormhole gate crossable in 4,488 sim-s (4.5 ms wall @ $1\times$) → dual cooldown adopted · $2\times v_{\text{esc}}$ at 10^{10} m from 50 M_SUN = 1152 km/s > 1000 km/s cap (clamp hierarchy clarified).

Prior rounds: E1 $\times 10$ column fix; E2 miscalibration (ϵ $10^8 \rightarrow 10^6$); E3 self-consistent r_s (2954.13 m from charter constants); E4 margin analysis; gravityforge existence re-verified by direct fetch $\times 3$ ("future-dated" accusation refuted: 2026-05-28 vs today 2026-08-31).

References: Google AI Studio Android constraints — ai.google.dev/gemini-api/docs/aistudio-android · Gaffer on Games, "Fix Your Timestep!" · symplectic-integrator literature (Verlet vs RK4 drift; Plummer softening) · PhET design principles & POE (Perkins et al. 2006; Adams et al. 2008) · Compose deferred state reads / draw-phase invalidation / `drawWithCache` — developer.android.com · Repositories evaluated, none imported: [igndev07/gravityforge](https://github.com/igndev07/gravityforge) · [KPhysics/KPhysics](https://github.com/KPhysics/KPhysics) · [joaomcl/boks2d](https://github.com/joaomcl/boks2d) · [ccampo133/NBodyJS](https://github.com/ccampo133/NBodyJS) · [AntonisPsarras/Aether-Gravity](https://github.com/Aether-Gravity/Aether-Gravity) · [alexvanyo/composelife](https://github.com/alexvanyo/composelife) · [jBox2D](https://github.com/jBox2D) · [libGDX/KTX](https://github.com/libGDX/KTX).

End of Roadmap v5.0. This document supersedes v2.0, v3.0, and v4.0. Say "Phase 0 approved" to receive the first Gemini-ready prompt.